EGAFNet: Edge-Guided Adaptive Fusion Network With Spatial–Frequency Interaction for Remote Sensing Change Detection

Peng Wang, Yangbin Liu, Quanqing Ma, Qingzhan Zhao, Zhiming Fang, and Yuchen Zheng, *Member, IEEE*

***Abstract*—Change detection (CD) is a pivotal technique for revealing surface differences within the same area across different time periods, serving as a crucial means for monitoring interactions between humans and nature. However, existing CD methods struggle to balance global semantics and local details across multilevel features, leading to blurred boundaries and missed detections within changed regions. Moreover, insufficient fusion of bitemporal features further results in pseudo changes in background areas. To address the aforementioned issues, this article proposes the edge-guided adaptive fusion network (EGAFNet) based on spatial–frequency interaction. Two parallel modules, namely the frequency attention module (FAM) and the multiscale perception module (MPM), are introduced to provide complementary enhancement of global and local information from the frequency domain and spatial multiscale perspectives, respectively. The adaptive decoupled fusion module (ADFM) is employed to progressively interact with bitemporal features, fully aligning spatiotemporal information and achieving deep fusion by leveraging multilevel attention. Subsequently, based on the interaction of parallel path features, the edge-driven wavelet reconstruction module (EWRM) is utilized to achieve fine-grained spatial frequency modeling of edge regions. Comparative experiments on three CD datasets (SYSU-CD, JLYH-CD, and SECOND-CD) demonstrate that the proposed method outperforms existing state-of-the-art methods, achieving $F1$-scores of 84.61%, 78.66%, and 73.01%, respectively. All codes are available at https://github.com/wpeng0166/EGAFNet**

***Index Terms*—Adaptive orthogonal fusion, change detection (CD), characteristics of multilevel features, edge-driven wavelet reconstruction, remote sensing (RS).**

I. Introduction

CHANGE detection (CD) is the process of identifying variations in scene elements from images acquired over the same geographical area at different temporal phases. Currently, CD has been successfully applied across diverse domains [1], [2], holding significant importance in land resource surveying [3], disaster assessment [4], vegetation dynamics monitoring [5], and urban planning [6].

Traditional remote sensing CD (RSCD) methods primarily rely on conventional image processing and computer vision techniques. Excluding manual visual interpretation, these approaches are broadly categorized into two classes: pixel-based CD (PBCD) [7], [8], [9], [10], [11], and object-based CD (OBCD) [12], constitute the main paradigms in early CD research. However, with the increasing level of detail in high-resolution remote sensing (RS) imagery, these traditional methods struggle to meet increasingly complex challenges.

In recent years, deep learning-based CD methods have demonstrated superior accuracy, global context awareness, and enhanced robustness compared to traditional methods, establishing themselves as the mainstream paradigm for CD tasks. Specifically, convolutional neural networks (CNNs) possess powerful feature extraction and self-learning capabilities, making them well-suited for CD [13], [14]. The introduction of Siamese networks fully exploits the deep feature (DF) correlations between bitemporal images, establishing this architecture as a benchmark in the RS field. Based on this architecture combined with U-Net, Caye Daudt et al. [15] develop the FC-Siam-Conc and FC-Siam-Diff networks. Similarly, Zhao et al. [16] utilize the pyramid pooling paradigm to design PSPNet, constructing a multilevel feature system that expands the scope of innovation in CD. The emergence of attention mechanisms has further improved model accuracy. Spatial attention [17], channel attention [18], and self-attention [19] have been widely applied, effectively enhancing feature representation capabilities. Although these attention modules improve performance, how to utilize attention mechanisms to adapt feature representations poses a new challenge in deep learning-based CD methods. Furthermore, since CNN models are inherently limited by local receptive fields and struggle to capture global features, transformer-based models equipped with global attention mechanisms have gained widespread popularity. Such as Changeformer [20], AGFormer [21]. These models excel in establishing long-range dependencies and overcoming the receptive field limitations of standard convolutions. Despite the excellent performance exhibited by such

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models, their high model complexity leads to a significant increase in the number of parameters, thereby exacerbating the memory burden.

Most existing model architectures adapt to the CD task in the form of an encoder, a bitemporal feature fusion module, and a decoder [22]. Some models attempt to further enhance the features extracted by the encoder before the fusion stage to make the attended features more discriminative. For instance, RFANet utilizes a feature enhancement module to uniformly aggregate multiscale features [23]. However, given the distinct characteristics of multilevel features, shallow features are rich in detailed representations, while DFs possess robust semantic information. The application of a uniform attention enhancement mechanism often leads to an imbalance between detailed and semantic features. In addition, the feature fusion stage is a critical component of the architecture [24], which evolves into diverse forms, such as the self-attention guided fusion in CFFormer [25] and the parallel attention fusion in MCECF [26]. Although these methods provide significant improvements over simple concatenation fusion, the fused features remain susceptible to background noise from bitemporal images, leading to pseudo changes and insufficient fusion degree, which makes it difficult to fully adapt to the decoder. In addition, the integrity of edge details is crucial for achieving fine-grained CD. ASGCN [27] utilizes graph convolutional networks to extract edge information, while PDANet [28] amplifies the sensitivity to edge information through the coordination of multiscale and gating mechanisms. Operational superposition solely in the spatial domain fails to distinguish real changes from external noise interference, leading to false detections and pseudo changes.

In contrast, the frequency domain enables targeted representation by decomposing high and low frequency components. Specifically, low-frequency features primarily carry structural semantics, while high-frequency features encapsulate details. This inherent spectral separation offers a superior solution to the imbalance between detail and semantic features, as it allows the network to maintain independent and sufficient focus on each spectral component, effectively avoiding the interference common in coupling. Several pioneering works have explored this direction; for instance, FTA-Net [29] proposes frequency-temporal awareness attention to enhance semantic focus, while DSFI-CD [30] utilizes spectral and spatial information to manipulate phase for improved discriminability. However, these methods often overlook the importance of cross-module synergistic interaction between spatial multiscale details and frequency information.

To address these limitations, we propose the edge-guided adaptive fusion network (EGAFNet) based on spatial–frequency interaction. This network aims to achieve efficient training while maintaining a relatively lightweight architecture, improving the fine-grained precision of edge textures, and effectively suppressing pseudo changes. Specifically, MobileNetV2 is utilized as the backbone encoder to efficiently extract multilevel features. Parallel paths comprising the frequency attention module (FAM) and multiscale perception module (MPM) are constructed to adapt to the distinct characteristics of multilevel features. The former utilizes the Fourier Transform to capture long-range global semantic dependencies in the frequency domain, while the latter exploits multiscale convolutions to mine local detail information in the spatial domain, thereby addressing the feature imbalance issue caused by single dimension enhancement. To address insufficient feature fusion, we propose an adaptive decoupled fusion module (ADFM). By incorporating the Gram–Schmidt orthogonalization principle [31], ADFM progressively aligns spatiotemporal information while suppressing noise through structured feature interactions. Finally, addressing the difficulty in expressing edge details, this article proposes the edge-driven wavelet reconstruction module (EWRM). This module utilizes the discrete wavelet transform (DWT) for frequency separation, realizing the cross-module interactive reconstruction between spatially refined DFs from the MPM and frequency-enhanced shallow features from the FAM. Combined with edge supervision, this module guides the network in constructing complete and accurate change boundaries.

In summary, the main contributions of this article are as follows.

1) We propose EGAFNet, a novel framework for RSCD based on spatial–frequency interaction. This method establishes a new mechanism for balancing semantics and details via dual spatial-frequency paths and leverages their cross-module synergistic interaction to precisely identify changed regions, thereby generating robust change masks.
2) We design parallel FAM and MPM paths to perform complementary enhancement targeting the distinct attributes of multilevel features, thereby improving the completeness of feature representation in the encoder.
3) An innovative fusion module, ADFM, is developed to realize multilevel progressive fusion of bitemporal features using orthogonal mapping and attention mechanisms, effectively eliminating background noise and enhancing the robustness of changed regions.
4) The EWRM is constructed under the guidance of explicit edge label supervision, utilizing the reconstruction mechanism of the DWT to achieve fine-grained modeling of change region boundaries.

The remainder of this article is organized as follows. Section II reviews related work. Section III describes the architecture of the proposed EGAFNet in detail. In Section IV, we analyze the experimental results and present ablation studies. Finally, Section V concludes our work.

II. Related Works

A. Feature Enhancement and Fusion

Deep learning-based frameworks have been widely adopted in RS, capable of autonomously learning high-level semantic information and capturing complex structural change information from images. Early CD networks are primarily built upon CNNs. Drawing inspiration from fully convolutional networks (FCNs) [32], early methods typically adopt a single encoder architecture, concatenating bitemporal images along the channel dimension as input to extract change features. To better recover spatial details, Peng et al. [33] design

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the UNet++ architecture, establishing the dominance of the "encoder–decoder" structure for end-to-end learning. Caye Daudt et al. [15] introduce Siamese networks into CD tasks, effectively avoiding the issue of unclear semantic representation caused by premature feature mixing. Zhang et al. [34] propose DSIFN, which introduces a deep supervision mechanism to improve the training efficiency of DFs and reduce information loss during upsampling. Compared to CNNs, which focus on local regions, Transformers can establish global pixel-wise associations. Bandara and Patel [20] design the ChangeFormer architecture, incorporating SegFormer [35] into the Siamese encoder to extract multilevel global semantic features.

Building on this, many studies have shifted toward feature enhancement, integrating attention mechanisms into Siamese network frameworks to improve CD accuracy. Feature enhancement is categorized into prefusion enhancement and postfusion enhancement. In terms of prefusion enhancement, Chen and Shi [36] propose STANet, which introduces a spatiotemporal attention module to capture global dependencies by calculating the correlation matrix between pixels. Cui et al. [37] utilize attention mechanisms to model temporal features, adaptively enhancing spatiotemporal feature representations to improve discriminability. In terms of postfusion enhancement, Fang et al. [38] propose SNUNet, combining dense connections with channel attention to significantly enhance the representation capability and localization accuracy of multiscale features. Su et al. [39] introduce a feature enhancer to enable the perception of complex dependencies and achieve interactive alignment of multiscale information.

As a crucial component of the network framework, feature fusion serves as a pivotal stage in CD tasks, aiming to integrate bitemporal features to highlight changed regions. Early feature fusion methods primarily rely on simple arithmetic operations, such as element-wise addition, subtraction, and channel concatenation. Although these methods are computationally efficient, they struggle to distinguish between change signals and background noise. Current models utilize more sophisticated mechanisms to fuse features. Liu et al. [40] combine channel cosine similarity with channel attention mechanisms for feature fusion, maintaining feature robustness while emphasizing distinctiveness. Zhou et al. [41] utilize spatial attention to dynamically adjust the weights of changed regions, improving the model's ability to detect subtle changes. Dong et al. [42] combine channel and spatial attention to capture multiscale dependencies across different dimensions, adaptively modeling the spatiotemporal relationships of bitemporal images. Wei et al. [43] employ a channel splitting strategy to interactively fuse bitemporal features, thereby reconstructing change clues between channels. Ying et al. [44] design a global–local dual branch fusion structure, which not only enhances global dependency modeling but also highlights local detail representation capabilities. Shen et al. [45] introduce an interactive self-attention mechanism to acquire holistic features in parallel from different perspectives and dimensions. Huo et al. [46] incorporate set concepts into feature fusion, flexibly constructing geometric-shaped neighborhoods for each pixel to accurately capture land cover information. Han et al. [47] utilize the bitemporal feature difference calculated by cosine similarity as prior information, serving as a query mechanism for interactive self-attention to accurately reconstruct changed regions in fused features. Wang et al. [48] design a three-feature fusion method integrating bitemporal features with features from the previous level, enabling the model to accommodate richer multilevel spatial information.

Although the above methods have achieved significant progress, limitations persist in current CD network frameworks. First, regarding feature enhancement, existing strategies often apply identical attention mechanisms across multiple feature levels. This single-dimensional strategy overlooks the distinct characteristics among multilevel features. Second, in terms of feature fusion, current methods often neglect the importance of deep decoupling interaction. Consequently, bitemporal features are difficult to fully fuse and align, frequently leading to the generation of pseudo changes in complex scenarios.

B. Edge Refinement

In complex RS imagery, the capability of models to capture edge details determines the precision of prediction results; thus, enhancing edge textures is essential. Edge refinement strategies are primarily categorized into two types: implicit edge guidance methods and explicit edge supervision methods.

Regarding implicit edge guidance methods, Chen et al. [49] propose EGDE-Net, which utilizes dilated convolutions to refine boundary information and embeds it into a Transformer encoder. Jiang and Wu [50] design LGCANet, employing parallel directional convolutions to guide the details of changed regions and enhance edge information processing. Liu et al. [51] propose EFICNN, which designs an edge guidance branch utilizing reverse attention and a Laplacian pyramid to focus on boundary information changes. Zhang et al. [52] introduce an edge refinement module, leveraging hierarchical features and local information to synergistically refine edge details. Regarding explicit edge supervision methods, Ma et al. [53] design EATDer, which achieves joint supervision of the output using edge labels and ground truth based on feature refinement, thereby highlighting changed regions. Bai et al. [54] develop an edge estimation module to predict edge information and guide change prediction. Zhu et al. [55] propose an edge perception module to obtain prior edge information, fusing it into subsequent features to enhance edge representation. Holail et al. [56] utilize an edge information change module to improve geometric accuracy and efficiently distinguish boundaries of changed regions. Li et al. [57] utilize features from different levels to perform semantic flow information fusion, enhancing the robustness of edge prediction.

Although the aforementioned methods have improved the accuracy of boundary information, the components for capturing boundary information mainly concentrate in the spatial domain with relatively simple structures, making it difficult to reconstruct fine-grained boundaries from blurred DFs. Therefore, introducing frequency domain tools such as the DWT for detailed edge reconstruction and leveraging the synergistic

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effect of both spatial and frequency domains is key to addressing this issue.

C. Deep Learning in the Frequency Domain

In RSCD, high-frequency and low-frequency components decoupled via frequency domain analysis are representative of details and structural semantics, respectively. Compared to the traditional spatial domain, the frequency domain serves as a pivotal tool for achieving effective feature decoupling and targeted representation. Qu et al. [58] propose DDNet, which is among the pioneering methods to integrate both frequency and spatial domain analysis into the CD task, thereby mitigating the influence of noise within feature representations. Zhou et al. [59] integrate the discrete cosine transform into JFSDNet to jointly leverage spatial and frequency domain information for capturing semantic structures, but the interaction between these two domains remains insufficient. Zhu et al. [29] propose the frequency-temporal fusion module in FTA-Net to enhance the interaction between the dual domains, thereby enabling more comprehensive information acquisition for the model. Li et al. [30] propose DSFI-CD, which innovatively applies diffusion models and frequency domain analysis to CD, utilizing diffusion processes to guide both spatial and frequency domains to generate more reliable prediction maps. Guo et al. [60] propose MSFNet, which utilizes the discrete Fourier transform to enhance the discriminative capability of differences between features. Huang et al. [61] integrate the fast Fourier transform (FFT) into the fine-tuning module to construct a more efficient frequency domain fine-tuning framework for the segment anything model. Song et al. [62] develop WaveHFG, which specifically employs high-frequency information as guidance to enhance the capability of the model to capture details. Recognizing the semantic discrepancy between abstract and shallow features, Xie et al. [63] propose FSG-Net, innovatively utilizing spatial-frequency synergistic gating to effectively capture semantic changes while suppressing noise interference. Wang et al. [64] develop RDSF-Net, which integrates wavelets into the Mamba architecture as an encoder to accurately extract ground object information from raw images. In multimodal CD, CFAT-MCD designed by Pu et al. [65] maps the informational advantages captured in the frequency domain into the spatial domain, thereby enhancing the spatial discriminability of multimodal features.

Although these methods explore diverse applications of the frequency domain, they predominantly rely on internal coupling during spatial and frequency interaction and lack comprehensive cross-module synergy. Moreover, designing frequency backbones across the entire pipeline incurs significant computational overhead. Consequently, maintaining long-range synergistic dynamic interaction between the dual domains has emerged as a critical issue for achieving a balance between semantic and detailed information.

III. Methodology

In this section, we first provide an overview of the proposed architecture, EGAFNet. Subsequently, the proposed MPM, FAM, ADFM, and EWRM are described in detail. Finally, the loss function is formulated to optimize the whole network.

A. Overview

The proposed EGAFNet is shown in Fig. 1. The proposed framework primarily consists of three stages: the feature extraction and enhancement stage (Stage 1), the feature fusion stage (Stage 2), and the upsampling and decoding stage (Stage 3).

In Stage 1, taking the bitemporal images as input, we utilize a weight sharing pretrained MobileNetV2 backbone to extract the initial hierarchical features, denoted as $\hat{\mathbf{F}}_i^t$, where $i \in \{1, 2, 3, 4\}$ and $t \in \{1, 2\}$ denote the feature level index and the temporal index, respectively. To address the imbalance between semantic and detailed information in multilevel features, we categorize them into shallow features ($i = 1, 2$) and DFs ($i = 3, 4$). We integrate FAM and MPM into this stage to construct the feature enhancement module. Specifically, FAM captures long-range dependencies in the frequency domain for the shallow features, while MPM enriches spatial multiscale details for the DFs. Consequently, the final enhanced multilevel bitemporal features are generated and denoted as $\mathbf{F}_i^t$.

In Stage 2, the extracted bitemporal features are processed by ADFM via a progressive fusion mechanism to generate the fused features $\mathbf{F}_i$, thereby fully extracting difference information and suppressing pseudo changes. Subsequently, the fused features $\mathbf{F}_1$ and $\mathbf{F}_4$ facilitate the cross-module interactive fusion of spatial and frequency domains within EWRM to generate an edge prediction map, which is supervised by $\mathbf{GT}_{edge}$. $\mathbf{GT}_{edge}$ is generated by applying the Sobel [57] operator to the original $\mathbf{GT}$, formulated as follows:

$$\mathbf{GT}_{edge} = \text{Sobel}(\mathbf{GT}) \tag{1}$$

where $\mathbf{GT}$ denotes the original ground truth, and $\mathbf{GT}_{edge}$ represents the edge ground truth. Subsequently, dual-domain strip attention mechanism (DSAM) [66] is utilized to fully fuse the original features with the edge information, generating edge-guided feature maps. In Stage 3, we employ the multiscale context aggregation module (MCAM) [67] as the decoder to perform upsampling and decoding, thereby generating the final prediction map.

B. Frequency Attention and MPMs

The hierarchical features encoded by the backbone contain rich feature information, and the properties of features vary across different levels. The shallow features exhibit rich detailed information, but their global semantic dependencies are insufficient. The DFs possess strong semantic information, yet they suffer from insufficient detail richness. As shown in Fig. 1, to address the deficiencies of both, we design the FAM specialized for the shallow feature branch and the MPM tailored for the DF branch. By jointly acting on the hierarchical features, they achieve complementary enhancement of semantics and details, thereby overcoming the noise problem caused by excessive discrepancies in feature-level properties.

1) Shallow Feature Branch: Shallow features possess a higher resolution relative to DFs; although they preserve more detailed information, their global perception capability is inferior to that of DFs. Despite the ability of self-attention

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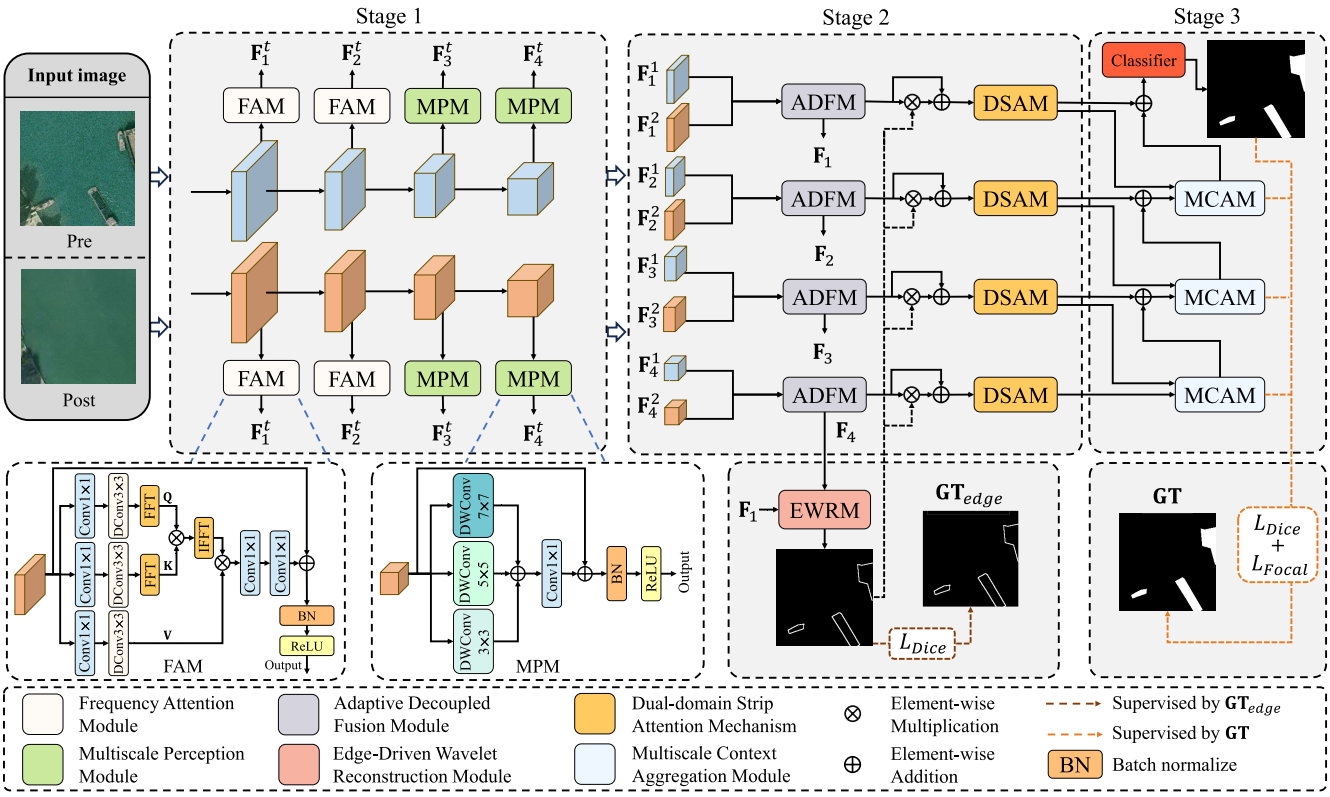


Fig. 1. Overview of the proposed EGAFNet. The features extracted by the backbone are complementarily enhanced by the integrated FAM and MPM. Subsequently, the ADFM fuses the bitemporal features by utilizing a progressive interactive decoupling mechanism. Meanwhile, the EWRM exploits the interaction between shallow frequency domain features and deep spatial domain features to generate edge prediction maps for guidance. The details of the proposed EGAFNet are elaborated in Section III.

mechanisms to focus on global semantic features, aggregating comprehensive semantic information remains challenging solely within the spatial domain, which can simultaneously lead to the loss of details. In the frequency domain, a single spectral point of the FFT aggregates information from all spatial pixels. Moreover, the parallel and decoupled processing of high-frequency and low-frequency components facilitates the preservation of details while simultaneously capturing a holistic perspective. Consequently, inspired by FSAS [68], we design a module that integrates FFT with the self-attention mechanism to acquire more comprehensive feature representations.

Specifically, the input feature $\hat{\mathbf{F}}_i^t$ is processed through three parallel branches. In each branch, a 1×1 convolution is employed to project the features, followed by a 3×3 dilated convolution to generate features with enlarged receptive fields, which serve as the Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$), respectively. Subsequently, $\mathbf{Q}$ and $\mathbf{K}$ are transformed into the frequency domain via FFT. The attention map is derived from the interaction between $\mathbf{Q}$ and $\mathbf{K}$ in the frequency domain, and then reverted to the spatial domain using inverse FFT (IFFT) to refine $\mathbf{V}$, generating the attention-weighted features $\mathbf{F}_{att}$. The formulation is as follows:

$$\mathbf{F}_{att} = \mathcal{F}^{-1}\left(\mathcal{F}\left(\mathbf{Q}\right) \otimes \mathcal{F}\left(\mathbf{K}\right)\right) \otimes \mathbf{V} \tag{2}$$

where $\otimes$ denotes the element-wise interaction operation. $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the FFT and IFFT operations. The weighted features subsequently undergo two consecutive 1×1 convolutions for further feature aggregation, yielding the aggregated features $\mathbf{F}_{agg}$

$$\mathbf{F}_{agg} = \mathrm{Conv}_{1\times1}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{F}_{att}\right)\right) \tag{3}$$

where $\mathrm{Conv}_{k\times k}(\cdot)$ denotes the $k \times k$ convolution operation. Finally, the aggregated features are added to the original input $\hat{\mathbf{F}}_i^t$ via a residual connection, and subsequently processed by batch normalization and ReLU activation to generate the final shallow features $\mathbf{F}_i^t$

$$\mathbf{F}_i^t = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathbf{F}_{agg} + \hat{\mathbf{F}}_i^t\right)\right) \tag{4}$$

where $\mathrm{BN}(\cdot)$ represents batch normalization and $\mathrm{ReLU}(\cdot)$ denotes the activation function.

2) DF Branch: Deep networks typically contain strong semantic information but lack detailed information. Therefore, the MPM is employed to enrich the spatial details of DFs by adopting parallel multiscale receptive fields to aggregate contextual information. Specifically, the input features $\hat{\mathbf{F}}_i^t$ are processed via three parallel depthwise separable convolutions with kernel sizes of 3×3, 5×5, and 7×7, and the outputs from these three branches are fused via element-wise addition to obtain the multiscale features $\mathbf{F}_{multi}$. This enables the network to perceive object information at varying scales without significantly increasing the computational burden, formulated as follows:

$$\mathbf{F}_{multi} = \mathrm{DWConv}_{3\times3}\left(\hat{\mathbf{F}}_i^t\right) + \mathrm{DWConv}_{5\times5}\left(\hat{\mathbf{F}}_i^t\right) + \mathrm{DWConv}_{7\times7}\left(\hat{\mathbf{F}}_i^t\right) \tag{5}$$

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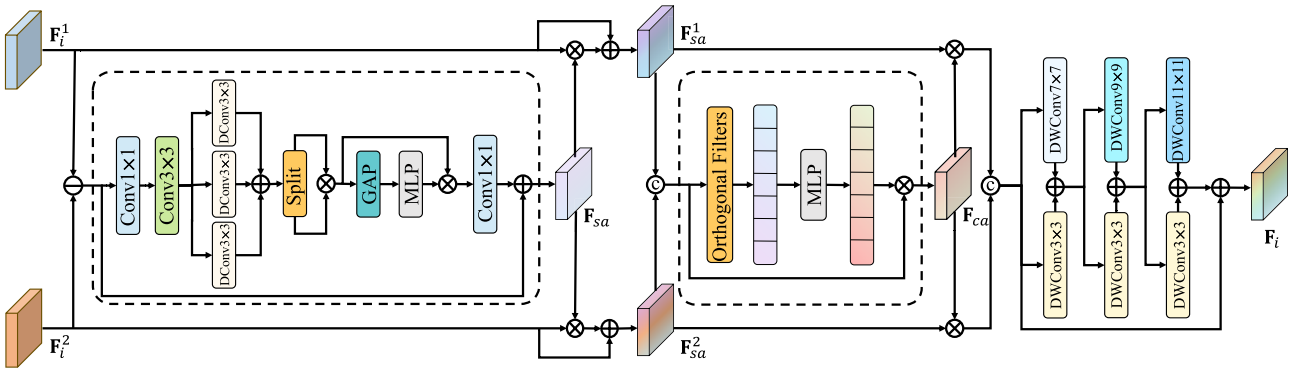


Fig. 2. ADFM enhances change information by utilizing spatial differences and suppresses pseudo changes via orthogonal channel attention. It employs convolutions with multiple scales for reconstruction to fully fuse the bitemporal features.

where $\mathrm{DWConv}_{k\times k}(\cdot)$ denotes the depthwise separable convolution with a kernel size of $k \times k$. Finally, a 1×1 convolution is employed to integrate channel information. The processed features are subsequently added to the original input $\hat{\mathbf{F}}_i^t$ via a residual connection, followed by batch normalization and activation, thereby generating the DFs $\mathbf{F}_i^t$. The formula is as follows:

$$\mathbf{F}_i^t = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{F}_{multi}\right) + \hat{\mathbf{F}}_i^t\right)\right). \tag{6}$$

C. Adaptive Decoupled Fusion Module

The objective of the feature fusion stage is to extract different information from semantically rich bitemporal features. As mentioned earlier, existing fusion strategies typically lack interactive guidance, causing change signals to be easily masked by the background. Furthermore, the reconstruction process of fused features is often overlooked, making it difficult to capture subtle changes. To this end, we propose the ADFM module as shown in Fig. 2, which fully fuses bitemporal features via progressive interactive decoupling and multiscale reconstruction.

To precisely capture change information, we first extract the difference features $\mathbf{F}_{diff}$ via element-wise subtraction. Subsequently, parallel dilated convolutions with varying dilation rates are employed to process and aggregate the difference cues. A gating mechanism based on channel splitting is utilized to adaptively suppress noise while amplifying the change signals. Following this, a channel attention mechanism is implemented using global average pooling (GAP) and a multilayer perceptron (MLP) to generate the difference-aware attention feature $\mathbf{F}_{sa}$. This feature interacts with the bitemporal features, respectively, to obtain the spatially enhanced features $\mathbf{F}_{sa}^1$ and $\mathbf{F}_{sa}^2$. The spatial interaction based on different information is expressed as

$$\mathbf{F}_{diff} = |\mathbf{F}_i^1 - \mathbf{F}_i^2| \tag{7}$$

$$\mathbf{F}_{mid} = \mathrm{Conv}_{3\times3}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{F}_{diff}\right)\right) \tag{8}$$

$$\mathbf{F}_{add} = \sum_{k=1}^{3} \mathrm{DConv}_{3\times3}^{k}\left(\mathbf{F}_{mid}\right) \tag{9}$$

$$\mathbf{X}_1, \mathbf{X}_2 = \mathrm{Split}\left(\mathbf{F}_{add}\right) \tag{10}$$

$$\mathbf{F}_{gate} = \mathbf{X}_1 \otimes \mathbf{X}_2 \tag{11}$$

$$\mathbf{F}_{se} = \mathbf{F}_{gate} \otimes \left(\mathrm{MLP}\left(\mathrm{GAP}\left(\mathbf{F}_{gate}\right)\right)\right) \tag{12}$$

$$\mathbf{F}_{sa} = \mathrm{Conv}_{1\times1}\left(\mathbf{F}_{se}\right) + \mathbf{F}_{diff} \tag{13}$$

$$\mathbf{F}_{sa}^t = \mathbf{F}_i^t + \mathbf{F}_i^t \otimes \mathbf{F}_{sa}, \quad t \in \{1, 2\} \tag{14}$$

where $|\cdot|$ denotes the element-wise absolute value operation. $\mathrm{DConv}_{3\times3}^{k}(\cdot)$ denotes the 3×3 dilated convolution in the kth branch, with dilation rates set to 1, 4, and 9, respectively.

The bitemporal features enhanced by different information are concatenated to generate $\mathbf{F}_{cat}$. To reduce background redundant information and thereby suppress pseudo changes, we argue that the standard GAP is insufficient, as it discards spatial distinctiveness. Therefore, an orthogonal mechanism [31] is introduced as a more robust alternative. The features are processed by a bank of orthonormal filters initialized via the Gram–Schmidt process to obtain $\mathbf{W}_{orth}$, followed by the squeeze and excitation process to generate the channel weights $\mathbf{F}_{ca}$. These weights interact with the bitemporal features again to enhance the discriminability of changed regions and the noise suppression capability in unchanged regions. Finally, the two branches are fused again to obtain the enhanced feature $\mathbf{F}_f$. The operation can be expressed as

$$\mathbf{F}_{cat} = \mathrm{Concat}\left(\mathbf{F}_{sa}^1, \mathbf{F}_{sa}^2\right) \tag{15}$$

$$\mathbf{W}_{orth} = \mathrm{Ortho}\left(\mathbf{F}_{cat}\right) \tag{16}$$

$$\mathbf{F}_{ca} = \mathrm{MLP}\left(\mathbf{W}_{orth}\right) \otimes \mathbf{F}_{cat} \tag{17}$$

$$\mathbf{F}_f = \mathrm{Concat}\left(\mathbf{F}_{sa}^1 \otimes \mathbf{F}_{ca}, \mathbf{F}_{sa}^2 \otimes \mathbf{F}_{ca}\right) \tag{18}$$

where $\mathrm{Concat}(\cdot)$ denotes the concatenation operation along the channel dimension. $\mathrm{Ortho}(\cdot)$ represents the orthogonal filter.

For the refined integration of change information in progressive interactive fusion features, $\mathbf{F}_f$ undergoes multiple reconstructions using a parallel addition of convolutions with large kernels for global context and convolutions with small kernels for local details. Subsequently, the final fused feature $\mathbf{F}_i$ is generated via a residual connection. The operation can be expressed as follows:

$$\mathbf{F}_{r1} = \mathrm{DWConv}_{3\times3}\left(\mathbf{F}_f\right) + \mathrm{DWConv}_{7\times7}\left(\mathbf{F}_f\right) \tag{19}$$

$$\mathbf{F}_{r2} = \mathrm{DWConv}_{3\times3}\left(\mathbf{F}_{r1}\right) + \mathrm{DWConv}_{9\times9}\left(\mathbf{F}_{r1}\right) \tag{20}$$

$$\mathbf{F}_{r3} = \mathrm{DWConv}_{3\times3}\left(\mathbf{F}_{r2}\right) + \mathrm{DWConv}_{11\times11}\left(\mathbf{F}_{r2}\right) \tag{21}$$

$$\mathbf{F}_i = \mathbf{F}_{r3} + \mathbf{F}_f, \quad i \in \{1, 2, 3, 4\} \tag{22}$$

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where $\mathbf{F}_{r1}$, $\mathbf{F}_{r2}$, and $\mathbf{F}_{r3}$ refer to the intermediate features generated at different reconstruction stages.

D. Edge-Driven Wavelet Reconstruction Module

The attention to edge information is positively correlated with the degree of prediction refinement. As stated previously, shallow features contain rich details while DFs contain strong semantic information. Many branches for edge assistance in existing research utilize the strengths of both to extract edge information via fusion. However, most designs for edge enhancement operate solely in the spatial domain, making it difficult to fuse details and semantic features precisely. These result in mutual interference between features, making it difficult to accurately predict boundary information. To overcome this limitation, we design EWRM, shown in Fig. 3, which enables the cross-module interactive fusion between shallow features $\mathbf{F}_1$ enhanced in the frequency domain and DFs $\mathbf{F}_4$ enhanced in the spatial domain. By separating information of high and low frequencies via DWT, it achieves precise fusion and reconstruction of semantic and detail information to generate edge prediction maps with high quality.

Unlike the selection of FFT for shallow features, DWT is chosen as the core frequency-domain tool for EWRM. In FFT, the basis functions extend infinitely across the entire spatial domain, which makes it difficult to fully decouple details from semantic information. Conversely, the basis functions of DWT are convergent, and each component only aggregates information from its corresponding window. The frequency components constructed by DWT achieve a more thorough decoupling compared to FFT. Therefore, integrating the low-frequency features isolated by DWT with semantic features can effectively suppress noise interference. In addition, our designed module aligns DFs with shallow features through upsampling, while the inherent downsampling characteristic of DWT provides a scale buffer, thereby reducing information loss caused by excessive upsampling. Ultimately, the choice of DWT is more conducive to the prediction of local edge information.

The complementarity between FFT and DWT establishes a synergistic advantage from global to local scales. FFT leverages its global information aggregation capability to capture macro contours for shallow features with high resolution, preventing the model from being misled by local details. DWT precisely isolates high and low frequency information and restores boundary blurring in DFs caused by upsampling at a local scale, which facilitates the reconstruction of more accurate edge details by the EWRM. Therefore, FFT and DWT are both integral components of the synergistic interaction within EGAFNet.

Specifically, the shallow feature $\mathbf{F}_1 \in \mathbb{R}^{H_s \times W_s \times C_s}$ is decomposed into a low frequency component **LL** and three high frequency components **LH**, **HL**, and **HH** via DWT. The formula is expressed as follows:

$$\{\mathbf{LL}, \mathbf{LH}, \mathbf{HL}, \mathbf{HH}\} = \mathrm{DWT}(\mathbf{F}_1) \tag{23}$$

where $\{\mathbf{LL}, \mathbf{LH}, \mathbf{HL}, \mathbf{HH}\} \in \mathbb{R}^{(H_s/2)\times(W_s/2)\times C_s}$, since the DWT operation halves the spatial resolution.

The low-frequency component is rich in semantic information, while the high-frequency components are rich in detailed information. The DF $\mathbf{F}_4 \in \mathbb{R}^{H_d \times W_d \times C_d}$ is unified to the same size as the low frequency feature **LL**. Then, the feature passes through depthwise convolution, normalization, and activation to obtain $\mathbf{F}_d$. Furthermore, the deep semantic weight map $\mathbf{W}_d$ is generated from $\mathbf{F}_d$ through a 1×1 convolution and Sigmoid activation. The process is as follows:

$$\mathbf{F}_4^{up} = \mathrm{UP}(\mathbf{F}_4) \tag{24}$$

$$\mathbf{F}_d = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{DWConv}_{3\times3}\left(\mathbf{F}_4^{up}\right)\right)\right) \tag{25}$$

$$\mathbf{W}_d = \mathrm{Sigmoid}\left(\mathrm{Conv}_{1\times1}(\mathbf{F}_d)\right) \tag{26}$$

where UP(·) denotes the upsampling operation. $\mathbf{F}_4^{up} \in \mathbb{R}^{(H_s/2)\times(W_s/2)\times C_d}$ represents the DF spatially aligned with the low frequency component, and $\mathbf{W}_d \in \mathbb{R}^{(H_s/2)\times(W_s/2)\times 1}$ denotes the generated semantic weight map. Sigmoid(·) represents the activation function.

The concatenation of $\mathbf{F}_d$ and **LL** undergoes channel integration and enhancement through convolution to obtain $\mathbf{F}_{deh}$

$$\mathbf{F}_{dll} = \mathrm{Concat}(\mathbf{LL}, \mathbf{F}_d) \tag{27}$$

$$\mathbf{F}_{deh} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Conv}_{1\times1}(\mathbf{F}_{dll})\right)\right)\right). \tag{28}$$

The high-frequency features and $\mathbf{F}_{deh}$ are concatenated, and the resulting features are enhanced using the Scharr operator to obtain $\mathbf{F}_{Scharr}$. In the process of decomposition and reconstruction, the Scharr operator is selected as the core tool for capturing enhanced boundary information. Compared to the Sobel operator used for generating $\mathbf{GT}_{edge}$, the Scharr operator significantly improves rotational invariance by optimizing weight ratios. This allows it to capture more subtle gradient changes and provide stronger response information within DFs. This aligns with the requirements of EWRM for fine-grained boundary modeling, enabling more effective guidance for the recovery of high-frequency components during wavelet reconstruction and thereby improving the accuracy of boundary prediction. The formula is expressed as follows:

$$\mathbf{F}_{all} = \mathrm{Conv}_{1\times1}(\mathrm{Concat}(\mathbf{F}_{deh}, \mathbf{LH}, \mathbf{HL}, \mathbf{HH})) \tag{29}$$

$$\mathbf{F}_{Scharr} = \mathbf{F}_{all} \otimes \mathrm{Scharr}\left(\mathrm{Conv}_{3\times3}(\mathbf{F}_{all})\right) + \mathbf{F}_{all} \tag{30}$$

where $\mathbf{F}_{all}$ is the concatenation of all frequency components. $\mathbf{W}_d$ is utilized to guide and enhance the semantic features of $\mathbf{F}_{Scharr}$, and $\mathbf{F}_{idwt}$ is reconstructed via inverse DWT (IDWT). Finally, the edge prediction map is optimized and output through depthwise convolutions with multiple receptive fields. The process is as follows:

$$\mathbf{F}_{idwt} = \mathrm{IDWT}(\mathbf{F}_{Scharr} \otimes \mathbf{W}_d) \tag{31}$$

$$\mathbf{F}_{d3} = \mathrm{DWConv}_{3\times3}(\mathbf{F}_{idwt}) \tag{32}$$

$$\mathbf{F}_{d5} = \mathrm{DWConv}_{5\times5}(\mathbf{F}_{d3}) \tag{33}$$

$$\mathbf{P}_{\mathrm{edge}} = \mathrm{DWConv}_{7\times7}(\mathbf{F}_{d5}) \tag{34}$$

where $\mathbf{F}_{d3}$ and $\mathbf{F}_{d5}$ represent the intermediate features, and $\mathbf{P}_{\mathrm{edge}}$ denotes the final output of the edge prediction.

E. Loss Function

In the training strategy, EGAFNet adopts a deep supervision mechanism to improve gradient propagation and enhance

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feature learning capabilities at different scales. The total loss function L_{total} consists of two parts: the auxiliary edge loss L_{edge} and the main CD loss L_{cd}.

Specifically, in this process, we utilize the Dice coefficient loss $\mathcal{L}_{dice}$ and the Focal loss $\mathcal{L}_{Focal}$. The specific expression is as follows:

$$\mathcal{L}_{dice} = 1 - \frac{2\sum_{i=1}^{N} y_i p_i + 1}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} p_i + 1} \tag{35}$$

$$\mathcal{L}_{Focal} = -\frac{1}{N}\sum_{i=1}^{N} \alpha\,(1-p_i)^{\gamma}\log(p_i) \tag{36}$$

where N denotes the total number of pixels in the ground truth, y_i and p_i represent the ground truth and the prediction results, respectively.

The pixel categories of changed regions and background regions in edge prediction are extremely imbalanced. Therefore, L_{edge} adopts the $\mathcal{L}_{dice}$ as the primary choice for the task of sparse change regions. L_{cd} uses a hybrid loss combining $\mathcal{L}_{Focal}$ and $\mathcal{L}_{dice}$, where the former focuses on subtle changes, while the latter ensures regional consistency. The formula is as follows:

$$L_{edge} = \mathcal{L}_{dice}\left(\mathbf{GT}_{edge}, \mathbf{P}_{edge}\right) \tag{37}$$

$$L_{cd} = \sum_{i=1}^{4} \left(\lambda \mathcal{L}_{Focal}\left(\mathbf{GT}, \mathbf{P}_i\right) + \mathcal{L}_{Dice}\left(\mathbf{GT}, \mathbf{P}_i\right)\right) \tag{38}$$

where $\mathbf{P}_i$ denotes the prediction result of the ith stage. Following the optimal configuration verified in the state-of-the-art method DMFANet [67], the hyperparameters α, γ, and λ are set to 0.25, 4, and 0.5, respectively. The configuration ensures the effective learning of foreground targets by suppressing the weights of irrelevant background regions while balancing the joint optimization of the loss functions. The final total loss is the sum of the two

$$L_{total} = L_{cd} + L_{edge}. \tag{39}$$

IV. Experiments

This section introduces three public CD datasets, implementation details, and evaluation metrics. Comparative experiments are conducted between EGAFNet and multiple methods. Furthermore, ablation studies are designed to verify the effectiveness and superiority of the proposed method.

Datasets: All experiment results are based on SYSU-CD [69], JLYH-CD, and SECOND-CD [70] datasets. The common characteristics of these datasets include diverse ground object element types and complex change forms, which increase the challenge of the CD task.

1) SYSU-CD is a high-resolution CD dataset featuring diverse ground object elements, showcasing diverse change scenarios such as urban buildings, suburban expansion, vegetation changes, road expansion, and marine engineering. The dataset contains 20 000 image pairs with a spatial resolution of 0.5 m, collected in Hong Kong between 2007 and 2014, and is divided into 256×256 pixel patches. We follow the official data split, where 12 000 image pairs are used for training, 4000 image pairs for validation, and 4000 image pairs for testing.
2) JLYH-CD includes 8000 image pairs with a spatial resolution of less than 0.75 m. It is a high-resolution CD dataset covering diverse changes in cultivated land, including multiple change forms where cultivated land transforms into various ground object elements and other elements transform into cultivated land. The data source is from the Jilin-1 Website of CGSTL, and is divided into 256×256 pixel patches. We follow the official data split, where 6000 pairs are used for training and 2000 pairs for testing.
3) SECOND-CD is a semantic CD dataset containing multiple land cover types. The main change elements include nonvegetated surfaces, trees, low vegetation, water bodies, buildings, and playgrounds, making the change forms complex and diverse. To obtain binary labels, all semantic change pixels are unified into a "change" class, while unchanged and background pixels are categorized as "no-change." The dataset contains 4662 image pairs, mainly from cities such as Hangzhou, Chengdu, and Shanghai, and each image is divided into 512×512 pixel patches. We follow the official data split, where 2968 image pairs are used for training and 1694 image pairs for testing.

Implementation Details: All our experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU. The proposed EGAFNet is implemented using the PyTorch framework. Data augmentation techniques, including image swapping, random rotation, flipping, cropping, and Gaussian blur, are employed. We initialize the learning rate to $5e^{-4}$ and select the AdamW optimizer to optimize the network. All datasets are trained for 100 epochs. Due to GPU memory limitations, the batch size is set to 16 for the SYSU-CD and JLYH-CD datasets, and four for the SECOND-CD dataset.

Evaluation Metrics: To measure the performance of the CD methods, five evaluation metrics are applied to our experiments. Specifically, the overall accuracy (OA) reflects the proportion of correctly predicted samples. The precision (Pre) represents the accuracy of all positive predictions. The recall (Rec) is used to evaluate the ratio of positive samples in all correct predictions. The intersection over union (IoU) denotes the degree of overlap between the predicted change area and the actual change area from the perspective of geometric distribution. The $F1$-score ($F1$) is an indicator that balances precision and recall. These metrics can be formulated as

$$\mathrm{F1} = \frac{2}{\mathrm{Pre}^{-1} + \mathrm{Rec}^{-1}} \tag{40}$$

$$\mathrm{Pre} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{41}$$

$$\mathrm{Rec} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{42}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FN} + \mathrm{TN} + \mathrm{FP}} \tag{43}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP}} \tag{44}$$

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TABLE I

COMPARISON OF CD TASKS ON TEST SETS OF SYSU-CD DATASETS BETWEEN THE PROPOSED EGAFNET AND OTHER STATE-OF-THE-ART METHODS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY. ASTERISKS (*) INDICATE MODELS WITHOUT PUBLICLY AVAILABLE CODE; THE RESULTS ARE CITED DIRECTLY FROM THEIR RESPECTIVE JOURNALS

Methods	Year	Pre	Rec	F1	IoU	OA
SNUNet [38]	2021	48.89	77.98	60.10	42.96	75.58
BIT [73]	2021	54.84	65.75	59.80	42.66	79.15
ChangeFormer [20]	2022	49.21	80.69	61.14	44.03	75.81
ChangeMamba [76]	2024	83.06	79.59	81.29	68.48	91.36
GaMPF [71]	2024	86.89	75.48	80.79	67.77	91.53
ScratchFormer [74]	2024	80.84	71.08	75.65	60.83	89.21
SFEARNet [57]	2025	84.87	81.25	83.02	70.97	92.16
AGformer [21]	2025	83.71	82.26	82.98	70.91	92.04
EFICNN [51]	2025	84.46	81.63	83.02	70.97	92.13
DMFANet [67]	2025	85.90	80.72	83.23	71.27	92.33
FTA-Net* [29]	2025	85.91	82.33	84.08	72.54	-
DSFI-CD* [30]	2025	84.46	89.23	85.94	75.35	93.90
HRMNet [72]	2025	86.90	78.29	82.37	70.02	92.10
HA2F [75]	2026	83.23	82.16	82.73	70.56	91.91
FILFBCD [61]	2026	85.15	82.48	83.80	72.11	92.23
Ours	-	85.79	83.47	84.61	73.33	92.84

where the TP, TN, FP, and FN stand for the corresponding totals of true positives, true negatives, false positives, and false negatives. Note that the $F1$ and IoU are the main metrics to evaluate the performance of the CD.

Comparison Methods: Fifteen representative CD methods are employed for comparison with the proposed method. The selected methods encompass five CNN-based methods, including SNUNet [38], GaMPF [71], EFICNN [51], FTA-Net [29], HRMNet [72], and DMFANet [67]; seven transformer-based methods, namely BIT [73], ChangeFormer [20], ScratchFormer [74], SFEARNet [57], DSFI-CD [30], HA2F [75], FILFBCD [61], and AGformer [21]; as well as one Mamba-based method, ChangeMamba [76]. Among them, EFICNN and SFEARNet are specifically focused on edge enhancement. Analogously, HRMNet implements a dynamic resupervision process targeting hard regions to refine detection results. FTA-Net, DSFI-CD, and FILFBCD adopt strategies for frequency enhancement. Furthermore, DSFI-CD incorporates diffusion models to diversify feature information. Similarly, HA2F and FILFBCD utilize DINO and SAM foundation models, respectively, to enhance their feature representation capabilities.

A. Results

In this section, we provide detailed experimental results of CD on three public datasets and compare the proposed EGAFNet with other state-of-the-art methods.

1) SYSU-CD: Comparative experiments are conducted on the SYSU dataset to evaluate the performance of the proposed EGAFNet against other state-of-the-art methods on the test set. The quantitative comparison results are presented in Table I, where the best results are highlighted in red, and the second-best results are in blue. Specifically, the proposed method achieves highly competitive performance in terms of Rec, $F1$, IoU, and OA, reaching 83.47%, 84.61%, 73.33%, and 92.84%, respectively. Compared with SFEARNet, which utilizes a similar edge auxiliary branch, the $F1$ and IoU scores of EGAFNet are improved by 1.59% and 2.36%, respectively. In comparison with similar models related to frequency, the $F1$ and IoU of our method increase by 0.53% and 0.79% over FTA-Net, and by 0.81% and 1.22% over FILFBCD, which employs foundation models. Furthermore, compared with DSFI-CD, a frequency model that utilizes diffusion models as an auxiliary, EGAFNet improves Pre by 1.33%. These experimental results fully demonstrate the superior performance of the proposed EGAFNet.

TABLE II

COMPARISON OF CD TASKS ON TEST SETS OF JLYH-CD DATASETS BETWEEN THE PROPOSED EGAFNET AND OTHER STATE-OF-THE-ART METHODS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Methods	Year	Pre	Rec	F1	IoU	OA
SNUNet [38]	2021	76.92	41.14	44.34	28.48	69.84
BIT [73]	2021	73.09	68.65	70.80	54.80	78.16
ChangeFormer [20]	2022	69.88	70.00	69.94	53.78	76.79
ChangeMamba [76]	2024	76.39	57.55	65.65	48.86	76.77
GaMPF [71]	2024	71.84	71.70	71.77	55.97	78.24
ScratchFormer [74]	2024	75.27	63.28	68.75	52.39	77.82
SFEARNet [57]	2025	74.34	66.86	70.40	54.32	78.31
AGformer [21]	2025	71.13	72.88	72.00	56.25	78.13
EFICNN [51]	2025	70.20	77.53	73.68	58.33	78.64
DMFANet [67]	2025	74.76	81.35	77.92	63.82	82.21
HRMNet [72]	2025	76.21	72.68	74.40	59.24	80.71
HA2F [75]	2026	71.89	66.55	69.11	52.81	77.06
FILFBCD [61]	2026	73.48	74.20	73.84	58.53	79.92
Ours	-	74.90	82.83	78.66	64.83	82.67

To provide a more intuitive understanding of the CD prediction, we visualize the results of the proposed EGAFNet and other state-of-the-art models on the SYSU-CD dataset, which contains diverse ground object changes. As shown in Fig. 4, the proposed EGAFNet demonstrates superior performance in suppressing pseudo changes and preserving edge detail accuracy. In the second and third rows, our method accurately identifies small target changes. In the fourth row, compared with other methods, EGAFNet effectively suppresses the interference noise in the top-left corner that other methods fail to eliminate. In the fifth and sixth rows, EGAFNet demonstrates excellent performance in identifying the integrity of large-scale changes. This fully demonstrates the effectiveness of our efforts in progressive fusion.

2) JLYH-CD: We conduct comparative experiments with other methods on the JLYH-CD dataset. The comparison results are shown in Table II. Specifically, the proposed method achieves the optimal results in Rec, $F1$, IoU, and OA, which are 82.83%, 78.66%, 64.83%, and 82.67%, respectively. Compared to the second-best metrics, Rec is improved by 1.48%, and IoU is improved by 1.01%, significantly outperforming other methods. Among them, the excellent Recall represents the stability of capturing change targets. Compared with EFICNN, which focuses on edge enhancement, the $F1$ is improved by 4.98%, and IoU is improved by 6.5%. In comparison with the latest frequency foundation model FILFBCD, the $F1$ and IoU are improved by 4.82% and 6.3%, respectively. In addition, although the latest HA2F also utilizes foundation models, EGAFNet outperforms it by 9.55% in $F1$ and 12.02% in IoU.

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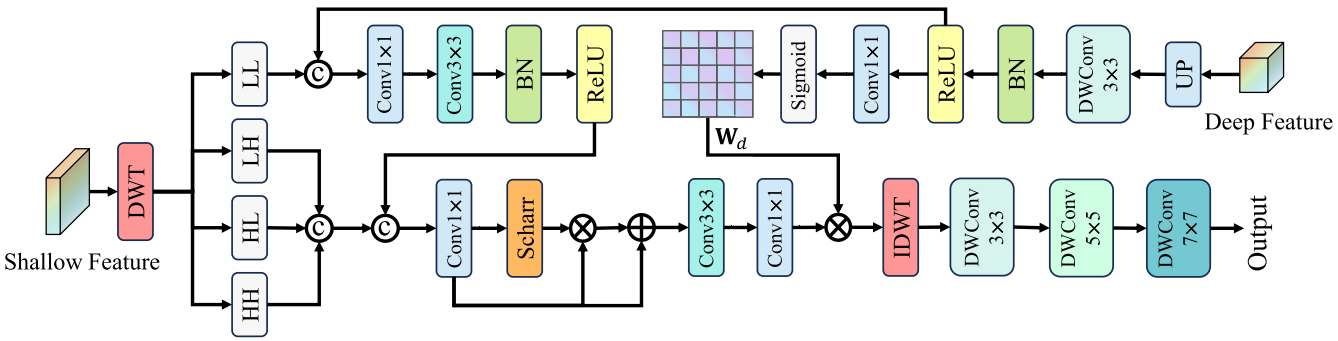


Fig. 3. EWRM achieves precise fusion of shallow and DFs in the frequency domain, enhances edge information using the Scharr operator followed by reconstruction, and utilizes multiscale receptive fields to generate a refined edge prediction map.

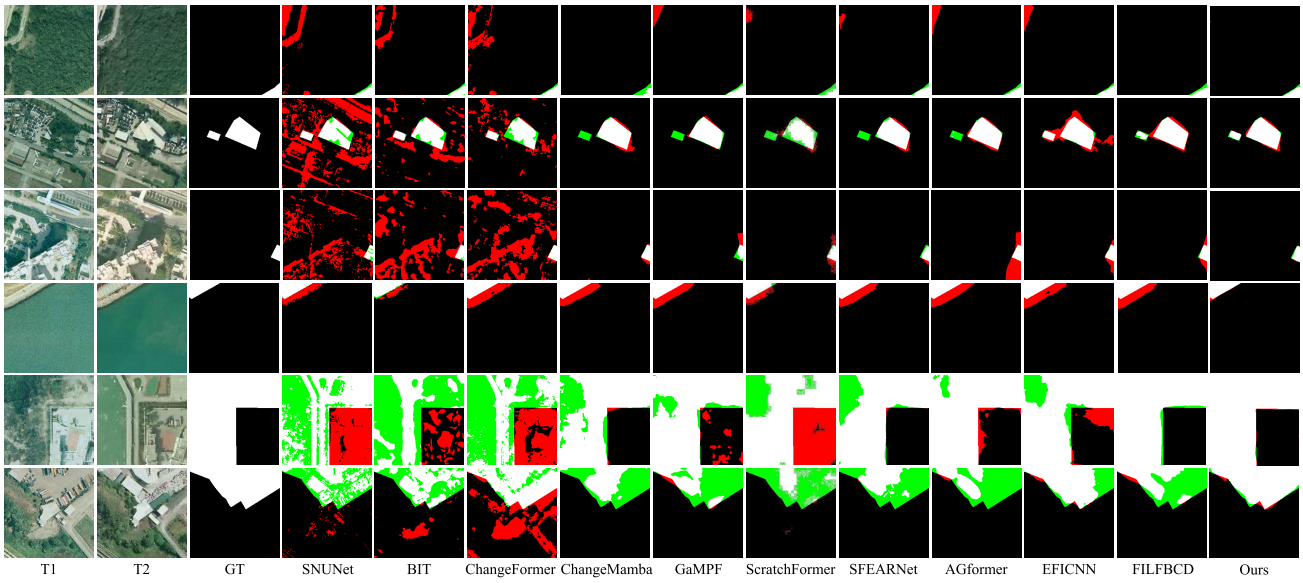


Fig. 4. Visualization results compared to other state-of-the-art methods on the test set of the SYSU-CD dataset. The red regions denote the false positives and the green regions are the false negatives.

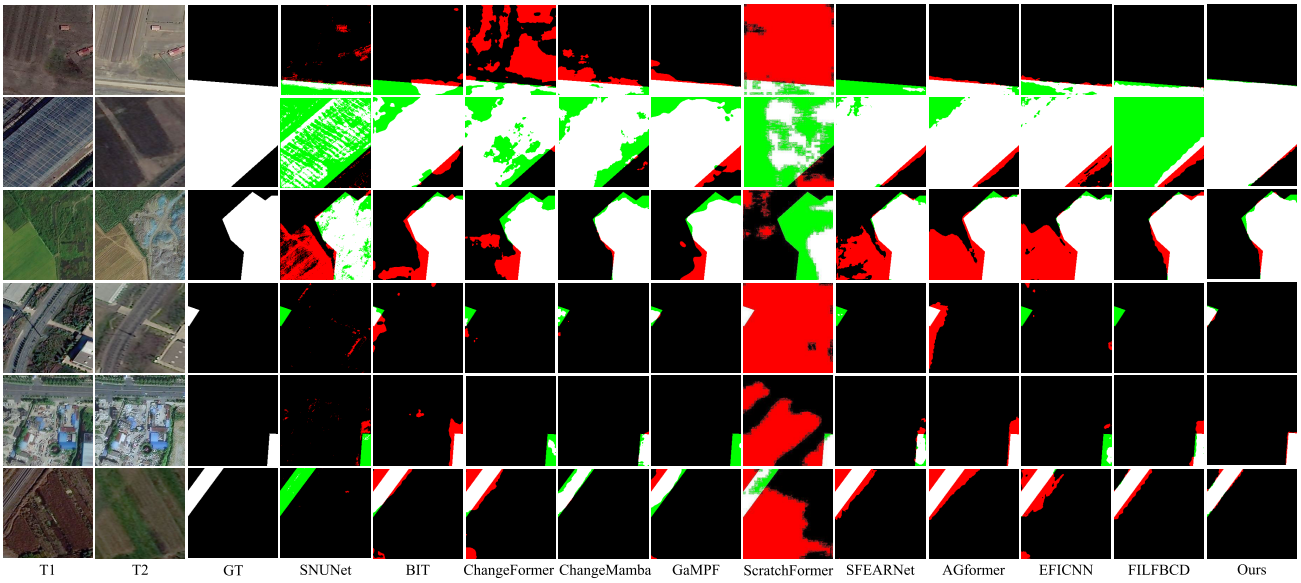


Fig. 5. Visualization results compared to other state-of-the-art methods on the test set of the JLYH-CD dataset. The red regions denote the false positives and the green regions are the false negatives.

The visualization results on the JLYH-CD dataset are shown in Fig. 5. The imaging styles of the bitemporal images in this dataset differ significantly, making them susceptible to interference from conditions other than the changed targets.

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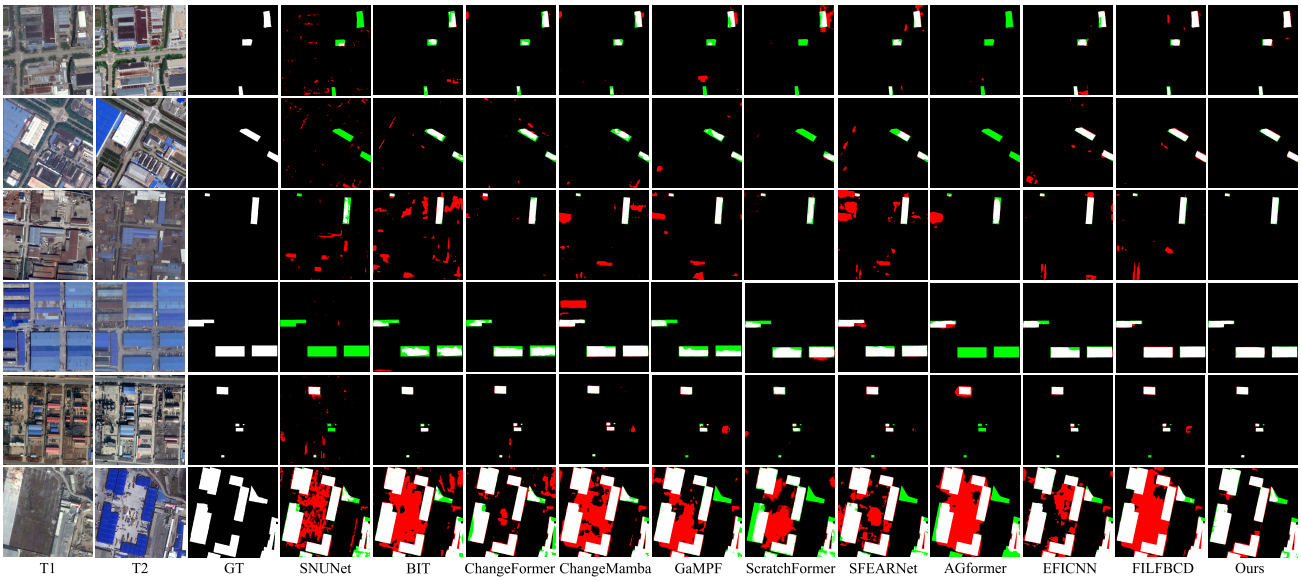


Fig. 6. Visualization results compared to other state-of-the-art methods on the test set of the SECOND-CD dataset. The red regions denote the false positives and the green regions are the false negatives.

TABLE III

COMPARISON OF CD TASKS ON TEST SETS OF SECOND-CD DATASETS BETWEEN THE PROPOSED EGAFNET AND OTHER STATE-OF-THE-ART METHODS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Methods	Year	Pre	Rec	F1	IoU	OA
SNUNet [38]	2021	63.11	33.71	43.94	28.16	82.74
BIT [73]	2021	74.36	63.52	68.52	52.11	88.29
ChangeFormer [20]	2022	74.74	68.02	71.23	55.31	88.97
ChangeMamba [76]	2024	69.88	70.22	70.05	53.91	87.95
GaMPF [71]	2024	70.32	65.66	67.91	51.41	87.55
ScratchFormer [74]	2024	62.59	73.31	67.53	50.98	87.92
SFEARNet [57]	2025	71.55	72.19	71.87	56.09	88.66
AGformer [21]	2025	71.48	50.06	58.88	41.73	85.97
EFICNN [51]	2025	72.59	70.91	71.74	55.93	88.79
DMFANet [67]	2025	74.44	70.68	72.46	56.81	89.23
HRMNet [72]	2025	71.74	72.23	72.04	56.29	88.73
HA2F [75]	2026	69.61	72.57	71.06	55.11	88.14
FILFBCD [61]	2026	72.77	71.69	72.23	56.53	88.77
Ours	-	75.71	70.50	73.01	57.49	89.54

Therefore, compared with other methods, EGAFNet demonstrates superior capability in capturing changed regions. In the second and third rows, our method performs well on large area changes, whereas other methods suffer from varying degrees of missed detections and false detections within the changed regions. In the fourth row, there is a significant resolution difference between the bitemporal images, leading to varying degrees of misjudgment in other methods, while EGAFNet stably detects the changed targets. This fully demonstrates the necessity of the complementary advantages of semantic features and detail features.

3) SECOND-CD: As shown in Table III, we present the performance comparison with other state-of-the-art methods on the SECOND-CD dataset. Specifically, our method achieves the optimal results in terms of Pre, $F1$, IoU, and OA, reaching 75.71%, 73.01%, 57.49%, and 89.54%, respectively. Compared to the second-best metrics, the $F1$ and IoU scores are improved by 0.55% and 0.68%, respectively. Since the image size of the SECOND-CD dataset is four times that of the previous two datasets, it demands a higher capability for global information modeling. Nevertheless, our method still outperforms transformer-based models that specialize in global context. For example, compared with AGformer, ScratchFormer, ChangeFormer, and FILFBCD, the $F1$ is improved by 14.13%, 5.48%, 1.78%, and 0.78%, respectively. In addition, compared with HRMNet, which focuses on dynamic resupervision for hard region mining, our method achieves further improvements of 0.97% in $F1$ and 1.20% in IoU.

The SECOND-CD dataset involves complex changes among various ground object elements, and the visualization results are shown in Fig. 6. In the first and second rows, EGAFNet accurately captures changed regions amidst complex land cover, whereas other methods suffer from varying degrees of missed detections and false detections. In the fourth and fifth rows, the edges of the changed elements detected by our method are almost consistent with the ground truth, demonstrating the effectiveness of the frequency edge auxiliary branch. In the sixth row, EGAFNet exhibits superior capability in suppressing background noise, while other methods show large-scale misjudgments, proving the strong anti-interference ability of our method.

B. Ablation and Analysis Studies

An ablation study is performed on the SYSU-CD dataset to investigate the proposed components of EGAFNet. In all experiments, the models are trained for 100 epochs with a batch size of 16.

1) Components: The results of the component ablation study are presented in Table IV. To verify the effectiveness of the proposed modules, we modify specific components within the baseline EGAFNet. First, we simultaneously replace the two feature enhancement modules, FAM and MPM, with

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TABLE IV
ABLATION STUDY OF EGAFNET ON SYSU-CD DATASET

Method	Dataset	Backbone	FAM	MPM	ADFM	EWRM	DSAM	MCAM	F1	IoU
EGAFNet	SYSU-CD	MobileNetV2	✗	✗	✓	✓	✓	✓	83.59	71.81
			✗	✓	✓	✓	✓	✓	83.80	72.11
			✓	✗	✓	✓	✓	✓	83.74	72.03
			✓	✓	✗	✓	✓	✓	83.01	70.95
			✓	✓	✓	✗	✓	✓	83.67	70.92
			✓	✓	✓	✓	✗	✓	83.90	72.26
			✓	✓	✓	✓	✓	✗	84.15	72.64
			✓	✓	✓	✓	✗	✗	83.79	72.11
			✓	✓	✓	✓	✓	✓	**84.61**	**73.33**

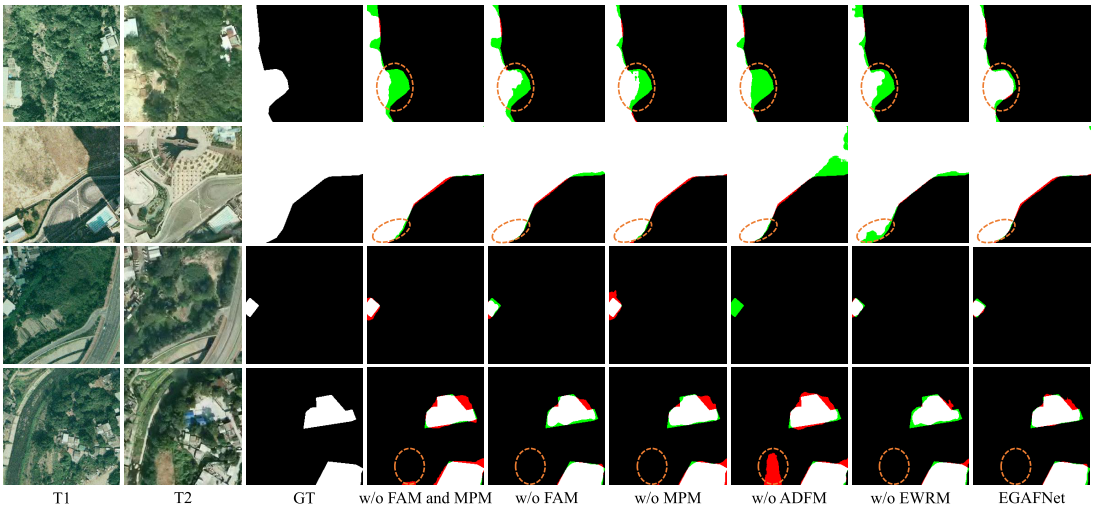


Fig. 7. Visualization results of the component ablation study on the test set of the SYSU-CD dataset. The red regions denote the false positives, and the green regions are the false negatives. The dashed circles indicate the highlighted regions.

1×1 standard convolutions, while retaining the normalization and activation layers. This modification results in a decrease of 1.02% in $F1$ and 1.52% in IoU. A similar replacement strategy is applied in the subsequent single-module ablations: when only FAM is replaced, $F1$ and IoU decrease by 0.81% and 1.22%, respectively; similarly, replacing MPM leads to drops of 0.87% in $F1$ and 1.30% in IoU. These results fully demonstrate the effectiveness of our strategy for the complementary enhancement of detail and semantic features. For the ablation of ADFM, we adopt the form of concatenation as a substitute, causing a significant decline in $F1$ and IoU of 1.60% and 2.38%, respectively. The results indicate that our proposed progressive fusion manner can fully explore the changed regions, thereby obtaining superior metrics. We use a traditional edge auxiliary branch to replace the EWRM module, i.e., upsampling DFs to the size of shallow features followed by concatenation for prediction, which leads to a relatively large IoU drop of 2.41%. This suggests that our proposed EWRM can accurately predict edge regions, avoiding negative effects caused by blurred edges.

In particular, additional ablation experiments are performed to isolate the respective roles of the auxiliary modules DSAM and MCAM within the framework. As shown in Table IV, the $F1$ and IoU still reach 83.90% and 72.26%, respectively, after the removal of DSAM. Regarding MCAM, when bilinear interpolation and 1×1 convolutions are adopted as dimensionality reduction substitutes, the $F1$ and IoU exhibit only slight decreases of 0.46% and 0.69%, respectively. The simultaneous removal of both modules results in a limited $F1$ decline of only 0.82%. It is noteworthy that the performance metrics after removing individual auxiliary components are consistently higher than the results obtained when the proposed core modules are missing. Although DSAM and MCAM provide beneficial effects, the fundamental performance gains and structural robustness of EGAFNet are primarily driven by the proposed spatial–frequency interaction and progressive fusion mechanisms.

Fig. 7 presents the visualization results of the component ablation study to verify the validity and reliability of our proposed modules. In the first row, only EGAFNet achieves complete predictions in the circled regions, whereas predictions from other variants with missing components remain incomplete, demonstrating the strong holistic synergy of our method. In addition, the predictions where only FAM or MPM is removed show better integrity than those where both are removed, indicating the necessity of these two modules. In the second row, the edge prediction accuracy without EWRM is significantly inferior to that of other experimental models. Regarding the removal of ADFM, small object changes in the third row become difficult to predict, while pseudo changes

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TABLE V
ABLATION STUDY OF FREQUENCY OPERATIONS IN FAM

Component	FFT	CA	SA	F1	IoU
FAM	✗	✓	✗	83.66	71.92
	✗	✗	✓	84.23	72.71
	✓	✗	✗	**84.61**	**73.33**

TABLE VI
INTERNAL ABLATION STUDY OF FREQUENCY COMPONENTS IN EWRM

Component	DWT	LL	HH/HL/LH	F1	IoU
EWRM	✗	✗	✗	83.45	71.60
	✓	✓	✗	83.93	72.31
	✓	✓	✓	**84.61**	**73.33**

TABLE VII
INTERNAL ABLATION STUDY OF ADFM COMPONENT

Component	Part 1	Part 2	Part 3	F1	IoU
ADFM	✓	✗	✗	83.93	72.31
	✓	✓	✗	84.14	72.62
	✓	✓	✓	**84.61**	**73.33**

TABLE VIII
INTERNAL ABLATION STUDY OF THE EWRM COMPONENT

Component	$\mathbf{W}_d$	Scharr	DF	F1	IoU
EWRM	✗	✗	✓	83.69	71.95
	✓	✗	✓	83.87	72.22
	✗	✓	✗	84.20	72.72
	✓	✓	✓	**84.61**	**73.33**

emerge in the fourth row due to noise interference, fully confirming the contribution of ADFM in highlighting changed regions and suppressing pseudo changes.

2) Analysis of the Frequency Operations: Frequency operations constitute an integral part of EGAFNet, and in this section, we investigate the ablation studies regarding frequency. As shown in Table V, we first present the ablation of FFT within the FAM. We disable the FFT and replace it with conventional channel attention and spatial attention, respectively, to compare with the full EGAFNet model that utilizes FFT. The results show that our method outperforms the variant using spatial attention by 0.95% in $F1$ and 1.41% in IoU. Compared with the variant using channel attention, the $F1$ and IoU scores are improved by 0.38% and 0.62%, respectively. This fully demonstrates the necessity of FFT within the designed FAM for capturing comprehensive information.

Table VI presents the ablation study of DWT within the EWRM. First, we evaluate the performance without DWT by utilizing only bilinear interpolation to replace all upsampling and downsampling processes related to DWT, while keeping other structures unchanged. As shown in the first row of Table VI, there is a significant performance gap between this variant and the original model, with differences of 1.16% in $F1$ and 1.73% in IoU. This verifies the importance of DWT in constructing frequency components. Subsequently, we utilize only the low-frequency component of DWT and set all high-frequency components to zero to prove the necessity of high-frequency components in edge recovery. The experiment demonstrates that the $F1$ and IoU decrease by 0.68% and 1.02%, respectively, indicating that high-frequency information is indispensable within the EWRM module.

3) Analysis of the ADFM: In this section, we investigate the internal ablation of the ADFM module to further validate the effectiveness of progressive fusion. As shown in Table VII, we denote the first differential interaction process as Part 1, the second process utilizing orthogonal attention interaction as Part 2, and the process employing parallel processing with large and small kernels after reconcatenation as Part 3. When utilizing only Part 1, the $F1$ and IoU reach 83.93% and 72.31%, respectively, outperforming the concatenation method described previously. When combining Part 1 and Part 2, the $F1$ and IoU improve by 0.21% and 0.31%, respectively, compared to using Part 1 alone. Employing the complete ADFM yields further improvements of 0.47% in $F1$ and 0.71% in IoU over the former configuration. The experiments demonstrate that our progressive fusion strategy is effective.

4) Analysis of the EWRM: In this section, we investigate the internal ablation of the EWRM module. As shown in Table VIII, we separately verify the effectiveness of the semantic spatial weight map $\mathbf{W}_d$ generated using DFs, the Scharr operator, and the DFs themselves within EWRM. Specifically, removing $\mathbf{W}_d$ and the Scharr operator results in an IoU improvement of 1.03% compared to the ordinary edge branch in Table IV, validating the effectiveness of introducing the frequency domain into the edge auxiliary branch. When removing only the Scharr operator while retaining $\mathbf{W}_d$, $F1$ and IoU increase by 0.18% and 0.27%, respectively, compared to removing both. Removing the DFs and their generated $\mathbf{W}_d$ yields an $F1$ of 84.20% and IoU of 72.72%, whereas our designed complete EWRM outperforms the configuration without DFs by 0.41% in $F1$ and 0.61% in IoU. The experiments fully demonstrate the necessity of the Scharr operator, DFs, and their corresponding $\mathbf{W}_d$.

Furthermore, Fig. 8 illustrates the visualization results of the EWRM ablation study across diverse scenarios to evaluate the effectiveness of the module in edge enhancement under different conditions. In the first and second rows, the variant without EWRM exhibits noticeable false positives and missed detections on small targets, while the complete EGAFNet identifies these tiny changes accurately. For the large area changes in the third and fourth rows, the absence of EWRM results in severe noise interference and incomplete detection. In the fifth row, when dual temporal images are affected by spectral interference resulting from lighting color differences, EGAFNet demonstrates superior boundary constraints under the edge guidance of EWRM. These visual comparisons confirm that EWRM is indispensable for maintaining structural integrity across different object scales and complex imaging environments.

5) Feature Visualization: Feature visualization intuitively validates the effectiveness of the proposed architectural design in balancing global semantics and local details across multilevel features. Fig. 9 displays the heatmaps of features from different modules using samples from the SYSU-CD dataset. In these maps, red regions indicate close atten-

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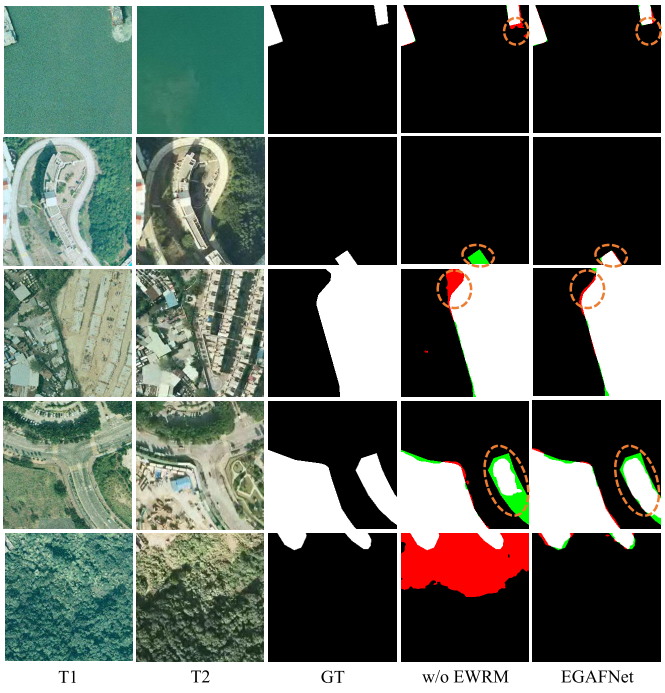


Fig. 8. Visualization results of the EWRM ablation study on the test set of the SYSU-CD dataset. The red regions denote the false positives, and the green regions are the false negatives. The dashed circles indicate the highlighted regions.

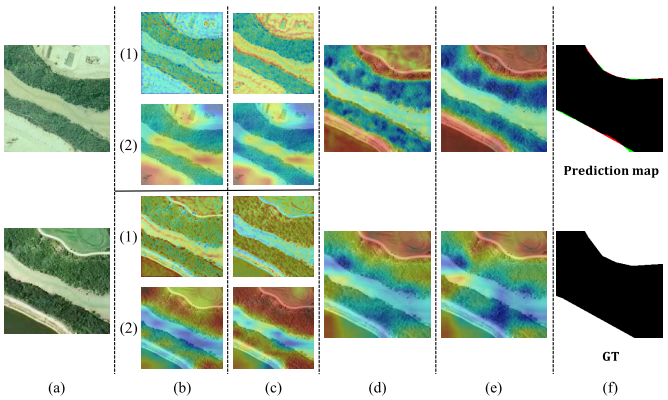


Fig. 9. Feature visualization in EGAFNet. (a) Input bitemporal images. (b) Features extracted by the backbone. (c) Features enhanced by parallel paths, where (1) and (2) are the features after FAM and MPM. (d) Features after ADFM. (e) Features after EWRM. (f) Prediction map and ground truth.

tion, while blue regions represent low attention. Among them, (1) and (2) denote shallow and DFs, respectively. For (d) and (e), the upper half represents shallow features while the lower half represents DFs. Comparing (b) and (c), it is evident that the parallel FAM and MPM effectively achieve complementary enhancement of features. Specifically, (1) demonstrates the global enhancement of FAM, which leverages the capability of FFT for global information aggregation to exhibit stronger global semantics compared to the shallow backbone features. (2) reveals the improved semantic representation and sensitivity to details of DFs via MPM. In the transition from (c) to (d), the progressive interactive fusion of ADFM allows both shallow and DFs to precisely

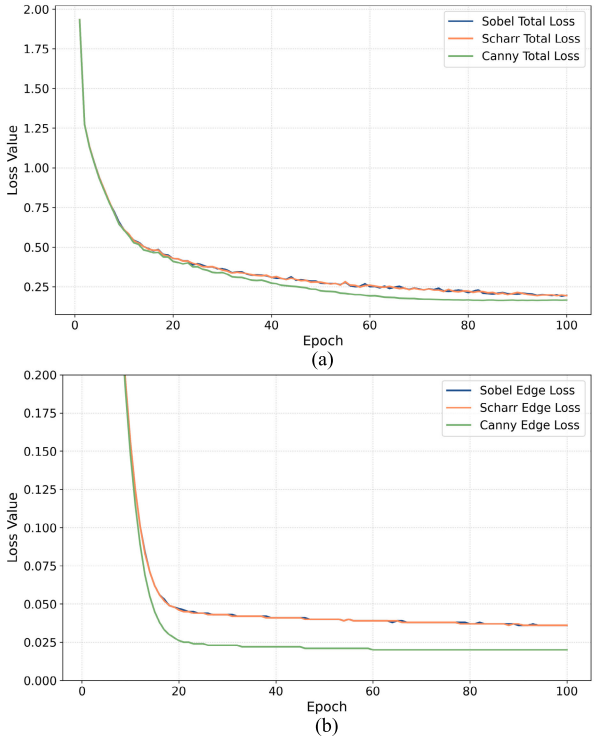


Fig. 10. Training loss convergence analysis on the SYSU dataset. (a) Total loss comparison under different edge operators. (b) Detailed edge loss convergence in the range [0.0, 0.2]. The curves demonstrate the training stability and robustness of our EGAFNet.

target changed regions. Notably, the heatmap of shallow features is highly consistent with the ground truth. Finally, from (d) to (e), the EWRM exhibits superior capabilities for capturing edges by interactively combining shallow features enhanced in the frequency domain with deep semantic features refined in the spatial domain. Compared to (d), (e) provides more accurate boundary details and stronger attention within changed regions, thereby generating a prediction map that almost matches the ground truth.

6) Analysis of the Edge Ground Truth: The $\mathbf{GT}_{edge}$ of the proposed model is generated using the Sobel operator. To investigate the training stability under different edge supervision strategies, we compare the Sobel operator with the Scharr and Canny operators. The original **GT** are, respectively, processed using Sobel, Scharr, and Canny edge operators to obtain the $\mathbf{GT}_{edge}$, which are then sequentially employed for the training of the proposed model. As illustrated in Fig. 10, (a) presents the total loss convergence curves of EGAFNet over 100 epochs, and (b) shows the edge loss convergence curves. The horizontal and vertical axes denote the epoch and the loss value, respectively. The blue, orange, and green lines represent the total and edge losses trained using $\mathbf{GT}_{edge}$ generated by Sobel, Scharr, and Canny operators, respectively.

Specifically, the various edge operators in both (a) and (b) exhibit favorable training convergence processes. Although the Scharr operator theoretically offers superior smoothness compared to the Sobel operator, their training trajectories almost overlap, indicating that its impact on stability is negligible. Furthermore, the convergence process of the Sobel

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TABLE IX

COMPARISON OF MODEL PARAMETERS AND EFFICIENCY ON THE SYSU-CD DATASET WITH 256 × 256 RESOLUTION. ASTERISKS (*) INDICATE MODELS WITHOUT PUBLICLY AVAILABLE CODE; PARAMS AND FLOPS ARE SOURCED FROM THE ORIGINAL PAPERS, WHILE INFERENCE LATENCY AND FPS ARE UNAVAILABLE

Method	Param (M)	FLOPs (G)	Infer (ms)	FPS
SNUNet [38]	3.01	11.73	9.61	276.63
BIT [73]	2.99	8.75	9.83	367.92
ChangeFormer [20]	13.94	5.83	10.15	437.89
ChangeMamba [76]	17.13	45.74	37.47	50.59
GaMPF [71]	69.64	15.09	8.97	182.76
ScratchFormer [74]	36.92	196.59	54.21	67.96
SFEARNet [57]	5.56	4.65	24.72	179.09
AGformer [21]	138.97	52.39	44.26	75.58
EFICNN [51]	21.32	29.02	18.81	91.60
DMFANet [67]	7.18	38.48	28.66	196.37
FTA-Net* [29]	6.71	4.93	-	-
DSFI-CD* [30]	10.14	22.28	-	-
HRMNet [72]	13.46	48.35	11.88	211.61
HA2F [75]	39.86	46.79	24.28	94.11
FILFBCD [61]	217.72	101.39	25.27	95.88
Ours	7.66	12.27	28.83	175.25

operator is relatively synchronized with that of the Canny operator, and no significant fluctuations are observed in subsequent epochs. Compared with the other two edge operators, the Sobel operator consistently maintains high convergence and stability, providing reliable and stable supervision signals for the proposed model.

7) Parameters and Efficiency: We evaluate the complexity of EGAFNet and other comparison methods on the SYSU dataset, with the results presented in Table IX. Infer denotes the inference latency for a single image pair, while FPS represents the throughput measured under a batch size of 16, expressed in image pairs per second.

Specifically, BIT and SFEARNet exhibit the lowest parameters and FLOPs, respectively, but their model performance is unsatisfactory, with $F1$ 24.81% and 1.59% lower than our method. ScratchFormer and AGformer possess the highest parameters and FLOPs, respectively, yet their performance remains inferior to ours. While DSFI-CD achieves a higher $F1$, its computational overhead is significantly greater than that of EGAFNet. Notably, while achieving a high $F1$ of 84.61%, EGAFNet maintains an inference latency of 28.83 ms, which is highly competitive. Furthermore, our model reaches a high throughput of 175.25 FPS, significantly faster than many recent models such as FILFBCD and EFICNN. In summary, the proposed EGAFNet achieves high performance results while maintaining an excellent balance between model complexity and practical execution efficiency.

V. CONCLUSION

In this article, we propose a novel CD method, termed EGAFNet, designed to address the imbalance between detailed features and semantic information, as well as the insufficiency of fine-grained edge details. Specifically, we introduce the FAM to enhance the long-range dependencies of shallow features and the MPM to capture detailed information within DFs, thereby achieving a complementary enhancement of semantic and detailed information. Subsequently, we propose the ADFM, which leverages different features and orthogonal attention to highlight changed regions and suppress pseudo changes, fully fusing bitemporal features through progressive interaction. Finally, we design the EWRM to facilitate the interactive fusion of frequency-enhanced shallow features and spatial DFs, generating edge information to guide the features for precise prediction. Extensive experiments conducted on three public datasets demonstrate both the effectiveness and efficiency of the proposed EGAFNet.

Limitations and Future Work: Due to the pursuit of model efficiency, the feature information extracted by the selected lightweight backbone is relatively limited, making it challenging to precisely capture change features when confronting limited samples among diverse ground objects.

In the future, we plan to integrate foundation models into this network, utilizing the combination of lightweight backbones and foundation models to extract richer features and enhance the generalization capability of the architecture. Meanwhile, we aim to design efficient and lightweight feature combination strategies, striving to maximize the integration of the advantages of both backbones.

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